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(1) Research Program/Project/Study Title: Vanilla Cookies enhanced with fresh camote leaves (Ipomoea batatas L.), malunggay leaves (Moringa oleifera), and lupu leaves (Alternanthera sessilis)	
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Academic Rank:	Instuctor I
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Sigma
Province:	Capiz
Region:	VI

(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☒ Basic	☐ Pilot Testing						
☐ Applied	☐ Tech. Promotion/Commercialization						

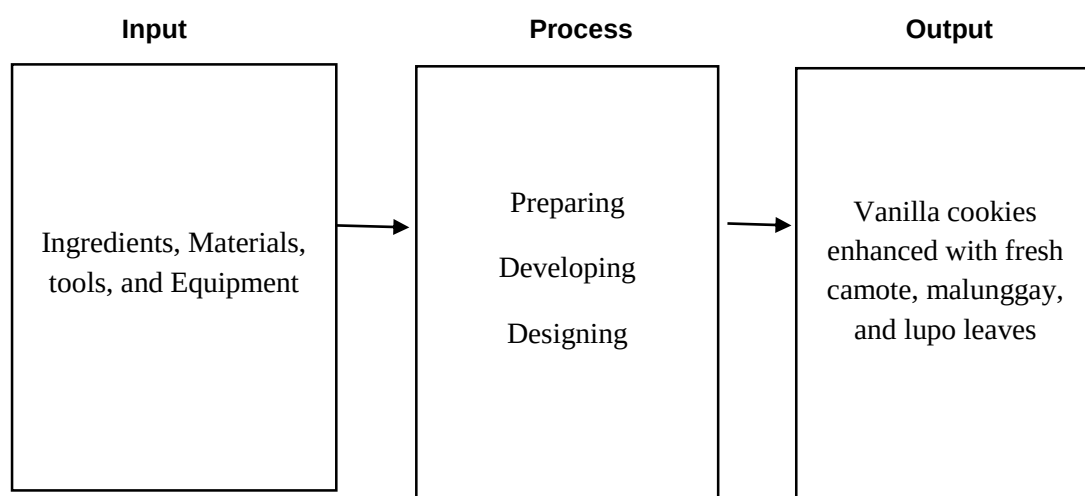
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(7) Sector/Commodity	(8) Discipline/Field of Specialization
(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.) Rationale Malnutrition in the Philippines causes a great deal of human suffering and is associated with more than half of all deaths of children worldwide. Malnutrition severely affects the socio-economic development of a nation because a work force that is stunted both mentally and physically may reduce work capacity (Dhakar et.al, 2011). “Modern nutritional science and in the context of food technology is providing more information on the functions and mechanisms of specific food components in health promotion and/or disease prevention” (Hsieh and Ofori, 2007). There are programs in the Philippines addressed to food insecurity through product development for alleviating malnutrition (Barba and Feliciano, 2002). Modern food technology thus provides an alternative health pathway for individuals who are unable to prepare their own healthy foods in order to conveniently obtain desired supplements or special nutrients from prepared baked products of their choice (Hsieh and Ofori, 2007). Greens encompass many medicinal and phytochemical effects, which are very useful in treating and preventing many diseases especially bioactive functional compounds like alkaloids, flavonoids, tannins and phenolic compounds (Sudha & Mathangi, 2013). Moringa leaves (<i>Moringa oleifera</i>) and “camote” leaves (<i>Ipomea batatas</i>) were added variables to baked product for good nutrition most especially among the diet of Filipinos nowadays. “Malunggay” is very essential for its medicinal value (Manaois et. al. 2013). Enhanced baked product with dried “malunggay” leaves powder, “camote” leaves decoction, and “camote” root could serve as a valuable source of nutrient in all age groups. Both “malunggay” and “camote” leaves are indigenous and inexpensive food ingredients to boost the nutritional quality of various baked products. In the Philippines, “Lupo” that commonly grows in marshy areas and wetlands. Locally, it is being used as vegetable and is claimed by folks to lower sugar and cholesterol levels and blood pressure. Despite its abundance and medicinal values, very few consume it due to its unacceptable taste. Food as Medicine” is one of the basic concepts of traditional Siddha Indian Medicine. Because the greens possess medicinal properties, Tamil culture usually include greens in household recipes, It is also known to be rich in antioxidants, sterol compounds which give cooling effect to eyes and body. It also relieves neuritis, treats 96 types of eye diseases, and aids disease free healthier life (Walter, Merish & Tamizhamuthu, 2014). With the above-mentioned scenario, the researchers wanted to determine the acceptability of vanilla cookies using camote, malunggay, and lupo leaves.	
(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)	
(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.) Objectives of the Study This study purposely to determine the acceptability of vanilla cookies enhanced with fresh camote, malunggay, and lupo leaves. Specifically;	

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1. determine the sensory qualities of vanilla cookies in terms of appearance, aroma, color, taste and texture among treatments;
2. determine the general acceptability of the product in terms of appearance, aroma, color, taste and texture among treatments;
3. determine if there is significant differences in the sensory qualities and general acceptability of the products among treatments,
4. determine the shelf life of the product in terms time and temperature (room and chilling temperature);
5. determine the moisture, ash and total fat content and microbial analysis of the best treatment.

(11) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.



(12) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Review of Related Literature

Malnutrition in the Philippines causes a great deal of human suffering and is associated with more than half of all deaths of children worldwide. Malnutrition severely affects the socio-economic development of a nation because a work force that is stunted both mentally and physically may reduce work capacity (Dhakar et.al, 2011).

Modern nutritional science and in the context of food technology is providing more information on the functions and mechanisms of specific food components in health promotion and/or disease prevention” (Hsieh and Ofori, 2007). There are programs in the Philippines addressed to food insecurity through product development for alleviating malnutrition (Barba and Feliciano, 2002). Modern food technology thus provides an alternative health pathway for individuals who are unable to prepare their own healthy foods in order to conveniently obtain desired supplements or special nutrients from prepared baked products of their choice (Hsieh and Ofori, 2007).

Moringa leaves (*Moringa oleifera*) and “camote” leaves (*Ipomea batatas*) were added variables to baked product for good nutrition most especially among the diet of Filipinos nowadays. “Malunggay” is very essential for its medicinal value (Manaois et. al. 2013). Enhanced baked product with dried “malunggay” leaves powder, “camote” leaves decoction, and “camote” root could serve as a valuable source of nutrient in all age groups. Both “malunggay” and “camote” leaves are indigenous and inexpensive food ingredients to boost the nutritional quality of various baked products.

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Thus, dried “malunggay” leaves powder and “camote” leaves decoction for pastry and filling products have a chance to prove in a commercial market that have great impact to our country's economy. It is within these contexts that the researcher is challenged to utilize malunggay and sweet potato leaves for dough and sweet potato root for fillings as a main ingredient in formulating pastry and fillings, (Aller et al. (2015) or starters, the leaves are an excellent source of antioxidants and contain high levels of vitamin A and C, as well as riboflavin, thiamin, folic acid, and niacin. Sweet potato vine leaves also provide impressive amounts of fiber, along with calcium, magnesium, manganese, zinc, copper, potassium, and iron, Dyer (2021).

Greens encompass many medicinal and phytochemical effects, which are very useful in treating and preventing many diseases especially bioactive functional compounds like alkaloids, flavonoids, tannins and phenolic compounds (Sudha & Mathangi, 2013).

In the Philippines, particularly in Iloilo it is commonly called “Lupo” that commonly grows in marshy areas and wetlands. Locally, it is being used as vegetable and is claimed by folks to lower sugar and cholesterol levels and blood pressure. Despite its abundance and medicinal values, very few consume it due to its unacceptable taste Food as Medicine” is one of the basic concepts of traditional Siddha Indian Medicine. Because the greens possess medicinal properties, Tamil culture usually include greens in household recipes, It is also known to be rich in antioxidants, sterol compounds which give cooling effect to eyes and body. It also relieves neuritis, treats 96 types of eye diseases, and aids disease-free healthier life (Walter, Merish & Tamizhamuthu, 2014).

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Methodology

This study will utilize the experimental research design to determine the sensory attributes and acceptability of vanilla cookies enhanced with fresh camote, malunggay, and lupo leaves in terms of appearance, aroma, color, taste, and texture. It will look into its shelf life in varying temperature and microbial analysis of the best treatment.

The experimental design to be use in the study will be the single factor experiment with Completely Randomized Design (CRD). In this study, the camote, malunggay, and lupo leaves will be added in different proportions; Treatment A – vanilla cookies (control), Treatment B – vanilla cookies with 10% camote leaves, Treatment C – vanilla cookies with 10% malunggay leaves, Treatment D – vanilla cookies with 10% lupo leaves, and Treatment E – vanilla cookies with 10% camote, 10% malunggay, and 10% lupo leaves.

Procedure in preparing the fresh camote, malunggay, and lupo leaves includes cleaning, washing, and draining.

Procedure in making vanilla cookies enhanced with camote, malunggay, and lupo leaves are the following: preparing the needed equipment, tools, and materials; sifting dry ingredients, measuring ingredients, mixing dry and wet ingredients, adding different proportions of fresh camote, malunggay, and lupo leaves, pouring batter in the baking sheet, baking, cooling, and packaging.

The arithmetic mean and Analysis of Variance (ANOVA) will be the statistical tools to be used in analyzing and interpreting the data. Using SPSS for data processing and analysis. The mean will be used to determine the appearance, aroma, color, taste and texture of the product. The ANOVA set at 0.01 alpha level will used to determine the significant differences on the sensory qualities of the products among treatments.

The study will used a scorecard as a research instrument. It looked into the five sensory qualities such as appearance, aroma, color, taste, and texture as the variables. These five sensory qualities will determine the general acceptability of vanilla cookies which could be the dependent variable.

The scorecard will employ a Nine-point Hedonic Scale wherein the respondents will be asked to rate each treatment classified into five categories. These catagories will be used in determining the sensory qualities of the product. Each quality will be assigned with adjectival description and it will be assigned with qualitative description such as 9 – Liked Extremely, 8 – Liked Very much, 7 – Liked

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Moderately, 6 – Liked Slightly, 5 – Neither Liked nor Disliked, 4 – Disliked Slightly, 3 – Disliked Moderately, 2 – Disliked Very Much, 1 – Disliked Extremely.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart. Below is an example. This can be modified base on your desired Work plan.)

Activities	July 2022	August 2022	September 2022	October 2022	November 2022	December 2022
Proposal/ Presentation						
Preparation of manuscript and Instrument						
Gathering of Data						
Data Encoding						
Interpretation of Data						
Presentation of Completed Research						

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

IEC materials, vanilla cookies enhanced with fresh camote, malunggay, and lupu leaves, brochures, and flyers

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

vanilla cookies enhanced with fresh camote, malunggay, and lupu leaves

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Children, teens, adults, and health conscious customers

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

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SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		2,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		3,000.00
- Meals		1,000.00
- Professional fees (laboratory analysis and other related fees)		7,000.00
- Miscellaneous (for incidental expenses)		1,000.00
- Printing		1,000.00
C. CAPITAL OUTLAY		
Total		15,000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Aller, M., Villarin, M., & Pascual, P. (2015). Product Development of Malunggay (Moringga oleifera) and Sweet Potato (Ipomea batatas) for Pastry and Fillings. Tropical Technology Journal, ISSN: 2199-8698, Volume 19, Issue 1, Article 5

Dyer, M. (2021). Potato Vine Plant Leaves: Are Sweet Potato Leaves Edible. <https://www.gardeningknowhow.com/edible/vegetables/sweet-potato/are-sweet-potato-leaves-edible.htm>

Hsieh and J.A. Ofori, (2007). Innovations in Food Technology for Health, Department of Nutrition, Food and Exercise Sciences," Florida State University, Tallahassee, Florida 32306-1493 Modern Asia Pacific Journal of Clinical Nutrition, 16 (Suppl 1):65-73, Review Article, 2007.

Sudha, K. & Mathangi, S. K. (2013). Functional compounds of some traditional greens and its medicinal properties. International Journal of Universal Pharmacy and Bio Sciences, 2(4), 2013

Walter, T.M., Merish, S., & Tamizhamuthu, M. (2014). Review of Alternanthera sessilis with reference to traditional Siddha medicine. International Journal of Pharmacognosy and Phytochemical Research, 6(2), 249-254.

(20) Capsule Curriculum Vitae (One-page brief C.V. only)